Complex Number

Mind map

Basics, powers of i

- $z = x + iy$, Re / Im
- $i^{4k+r} = i^r$
- Equality: parts equal

Algebra, square root

- Divide: multiply by conjugate
- $\sqrt{a + ib} = x + iy$, $x^2 + y^2 = |z|$
- Equations with $\bar{z}$: modulus first

Conjugate

- Mirror in real axis
- Real: $z = \bar{z}$; imaginary: $z = -\bar{z}$

Modulus

- $z\bar{z} = |z|^2$
- $|z_1 z_2| = |z_1||z_2|$
- Triangle inequality, equality = collinear

Argument

- Principal value in $(-\pi, \pi]$
- Product $\to$ sum, quotient $\to$ difference

Polar / exponential form

- $re^{i\theta} = r\,\mathrm{CiS}\,\theta$
- $1 + e^{i\theta} = 2\cos\frac{\theta}{2}e^{i\theta/2}$
- $|z| = 1$: put $z = e^{i\theta}$

Geometry

- $|z_1 - z_2|$ = distance
- Section formula, G, I, H
- $0, z_1, z_2, z_1 + z_2$ parallelogram

Loci

- Circle, bisector, Apollonius
- Ellipse / hyperbola by sums / differences
- Else put $z = x + iy$

Angles

- $\arg\frac{z_3 - z_1}{z_2 - z_1}$ = angle at z_1
- Real ratio: parallel; imaginary: perpendicular
- Fixed angle: arc of circle

Rotation

- Multiply by $e^{i\theta}$
- Equilateral: $\sum z_1^2 = \sum z_1 z_2$
- Regular polygon vertices

De Moivre, nth roots

- $(e^{i\theta})^n = e^{in\theta}$
- Roots: use $\theta + 2k\pi$
- Real root: $\mathrm{Re} = \mathrm{Im} = 0$

Cube roots of unity

- $1 + \omega + \omega^2 = 0$, $\omega^3 = 1$
- Powers mod 3
- $x^2 \pm x + 1$ roots

nth roots of unity

- Sum 0, product $(-1)^{n-1}$
- Regular n-gon on $|z| = 1$
- $x^n - 1 = \prod(x - \alpha_k)$

Series, binomial

- $C + iS$ as GP
- Put $x = i, \omega$ in $(1 + x)^n$

Line and circle

- $\bar{a}z + a\bar{z} + b = 0$
- $z\bar{z} + \bar{a}z + a\bar{z} + b = 0$
- Concyclic: cross ratio real